Random Samples, Estimation, Confidence Intervals, and Intro to Hypothesis Testing

Critical-value convention

The lecture notes use **upper-tail notation**:

$$P(Z > z_\alpha) = \alpha, \qquad P(T_\nu > t_{\alpha,\nu}) = \alpha, \qquad P(\chi^2_\nu > \chi^2_{\alpha,\nu}) = \alpha.$$

Therefore:

- a two-sided level-α procedure uses $z_{\alpha/2}$ or $t_{\alpha/2,\nu}$;
- a one-sided level-α procedure uses z_α or $t_{\alpha,\nu}$.

---

1. Random Samples and Statistics

1.1 Random sample

A random sample of size n from a population distribution is

$$X_1, \ldots, X_n,$$

where the random variables are independent and identically distributed. We write

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F.$$

Before observation, X_i is a random variable. After observation, its numerical value is written as x_i.

1.2 Statistic

A statistic is a function of the sample that contains no unknown parameter:

$$T = T(X_1, \ldots, X_n).$$

Examples:

$$\bar{X}, \qquad S^2, \qquad \sum_{i=1}^n X_i, \qquad \max_i X_i.$$

If μ is unknown, then $X_1 - \mu$ is not a statistic.

1.3 Sample mean

$$\boxed{\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i}$$

For observed data,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i.$$

1.4 Sample variance and standard deviation

$$\boxed{S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\boxed{S = \sqrt{S^2}}$$

For observed data, use s^2 and s.

Useful computational formulas:

$$\boxed{S^2 = \frac{1}{n-1}\left(\sum_{i=1}^n X_i^2 - n\bar{X}^2\right)}$$

and

$$\boxed{S^2 = \frac{n\sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i\right)^2}{n(n-1)}}.$$

---

2. Point Estimation

2.1 Estimator and estimate

Let θ be an unknown parameter. An estimator is a statistic

$$\hat{\theta} = T(X_1, \ldots, X_n).$$

After the sample is observed, the numerical value is called an estimate.

2.2 Bias and mean squared error

$$\boxed{\operatorname{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta}$$

The estimator is unbiased if

$$\boxed{E[\hat{\theta}] = \theta}.$$

The mean squared error is

$$\boxed{\operatorname{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]}$$

and

$$\boxed{\operatorname{MSE}(\hat{\theta}) = \operatorname{Var}(\hat{\theta}) + \operatorname{Bias}(\hat{\theta})^2}.$$

2.3 Properties of the sample mean

Assume

$$E[X_i] = \mu, \qquad \operatorname{Var}(X_i) = \sigma^2.$$

Then

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \mu.$$

Hence

$$\boxed{E[\bar{X}] = \mu}.$$

If the observations are independent,

$$\operatorname{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n} \operatorname{Var}(X_i) = \frac{\sigma^2}{n}.$$

Therefore

$$\boxed{\operatorname{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}}.$$

When σ is unknown, estimate the standard error by

$$\boxed{\widehat{\operatorname{SE}}(\bar{X}) = \frac{S}{\sqrt{n}}}.$$

Here S is the sample standard deviation, not the standard error itself.

2.4 Properties of the sample variance

For an i.i.d. sample with variance σ^2,

$$\boxed{E[S^2] = \sigma^2}.$$

A useful identity is

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations gives

$$E\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = (n-1)\sigma^2.$$

Thus division by $n - 1$ makes S^2 unbiased for σ^2.

However,

$$\boxed{S \text{ is generally not unbiased for } \sigma.}$$

3. Method of Moments

3.1 Sample moments

The k-th population moment is $E[X^k]$. The corresponding sample moment is

$$\boxed{M_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k}.$$

In particular,

$$M_1 = \bar{X}, \qquad M_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

Also,

$$M_2 - M_1^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{n-1}{n}S^2.$$

Therefore

$$M_2 - M_1^2 \neq S^2.$$

3.2 Standard exam procedure

Assume that the required population moments exist and that the moment equations identify the unknown parameters uniquely. If the model has p unknown parameters:

1. Write the first p population moments in terms of the parameters.
2. Replace the population moments by $M_1, \dots, M_p$.
3. Solve the resulting equations.
4. Check that the answers lie in the parameter space.

3.3 Binomial example

Suppose

$$X_i \sim \text{Binomial}(N, p),$$

where N is known and p is unknown. Then

$$E[X] = Np.$$

Set

$$\bar{X} = N\widehat{p}.$$

Therefore

$$\boxed{\widehat{p}_{\text{MOM}} = \frac{\bar{X}}{N}}.$$

3.4 Gamma example

Use the shape-scale parameterization

$$X \sim \text{Gamma}(\alpha, \beta),$$

with

$$E[X] = \alpha\beta, \qquad \text{Var}(X) = \alpha\beta^2.$$

The moment equations are

$$M_1 = \widehat{\alpha}\widehat{\beta},$$

$$M_2 - M_1^2 = \widehat{\alpha}\widehat{\beta}^2.$$

Hence

$$\boxed{\widehat{\beta} = \frac{M_2 - M_1^2}{M_1}}$$

and

$$\boxed{\widehat{\alpha} = \frac{M_1^2}{M_2 - M_1^2}}.$$

Always check whether the second Gamma parameter is a scale or a rate.

4. Maximum Likelihood Estimation

4.1 Likelihood and log-likelihood

For an i.i.d. sample with density or probability mass function $f(x \mid \theta)$, the likelihood is

$$\boxed{L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)}.$$

The data are fixed and θ varies.

Define the log-likelihood

$$\boxed{\ell(\theta) = \log L(\theta)}.$$

Since $\log$ is increasing, maximizing L is equivalent to maximizing ℓ.

4.2 Complete MLE solution procedure

Use the following steps in an exam.

Step 1: State the parameter space

Write the allowed values of the parameter, for example

$$\lambda \geq 0, \qquad \sigma^2 > 0, \qquad 0 < p < 1.$$

Step 2: Write the likelihood

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

Use independence to form the product.

Step 3: Write the log-likelihood

$$\ell(\theta) = \log L(\theta).$$

Remove additive constants that do not depend on θ if this simplifies the calculation.

Step 4: Solve the score equation

$$\ell'(\theta) = 0.$$

This step gives only a **candidate** maximizer.

Step 5: Prove that the candidate is a global maximum

The condition

$$\ell''(\hat{\theta}) < 0$$

shows only that $\hat{\theta}$ is a strict local maximum. To conclude that it is the MLE, also justify that it is the global maximizer. Common methods are:

- show that $\ell''(\theta) < 0$ for every allowed θ, so ℓ is strictly concave;
- show that $\ell'(\theta) > 0$ before $\hat{\theta}$ and $\ell'(\theta) < 0$ after it;
- compare all stationary points and boundary values;
- show directly that the likelihood is largest at the candidate.

Step 6: Check boundaries and the parameter space

A maximum may occur at a boundary even when $\ell'(\theta) = 0$ has no solution. The final answer must belong to the parameter space.

4.3 Poisson MLE: complete derivation

Suppose

$$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda), \qquad \lambda \geq 0.$$

The probability mass function is

$$f(x_i \mid \lambda) = e^{-\lambda} \frac{\lambda^{x_i}}{x_i!}.$$

Likelihood

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \\ &= \frac{e^{-n\lambda} \lambda^{\sum_i x_i}}{\prod_i x_i!}. \end{aligned}$$

Log-likelihood

$$\ell(\lambda) = -n\lambda + \left(\sum_i x_i \right) \log \lambda - \sum_i \log(x_i!).$$

Score equation

$$\ell'(\lambda) = -n + \frac{\sum_i x_i}{\lambda}.$$

Set $\ell'(\lambda) = 0$:

$$-n + \frac{\sum_i x_i}{\lambda} = 0.$$

Hence

$$\widehat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{x}.$$

Maximum check

$$\ell''(\lambda) = -\frac{\sum_i x_i}{\lambda^2}.$$

If $\sum_i x_i > 0$, then

$$\ell''(\lambda) < 0$$

for all $\lambda > 0$. Therefore ℓ is strictly concave and

$$\boxed{\widehat{\lambda}_{\text{MLE}} = \bar{X}}.$$

If every observation is zero, the likelihood is decreasing in λ, so the maximum is at the boundary $\widehat{\lambda} = 0$, when 0 is allowed.

4.4 Normal MLE: complete derivation

Suppose

$$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2), \qquad \sigma^2 > 0.$$

Let $v = \sigma^2$. Then

$$f(x_i \mid \mu, v) = \frac{1}{\sqrt{2\pi v}} \exp\left[-\frac{(x_i - \mu)^2}{2v} \right].$$

The log-likelihood is

$$\ell(\mu, v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{1}{2v} \sum_{i=1}^{n} (x_i - \mu)^2.$$

Estimate μ

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v}\sum_{i=1}^{n}(x_i - \mu).$$

Set it equal to zero:

$$\sum_{i=1}^{n}(x_i - \mu) = 0.$$

Therefore

$$\widehat{\mu} = \bar{x}.$$

The second derivative is

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{v} < 0.$$

Thus, for fixed v, the log-likelihood is maximized at $\mu = \bar{x}$.

Estimate $v = \sigma^2$

Substitute $\mu = \bar{x}$. Let

$$Q = \sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The profile log-likelihood is

$$\ell(v) = C - \frac{n}{2}\log v - \frac{Q}{2v},$$

where C does not depend on v.

Differentiate:

$$\ell'(v) = -\frac{n}{2v} + \frac{Q}{2v^2}.$$

Set $\ell'(v) = 0$:

$$-nv + Q = 0.$$

Hence

$$\hat{v} = \frac{Q}{n}.$$

Now

$$\ell''(v) = \frac{n}{2v^2} - \frac{Q}{v^3}.$$

Assume $Q > 0$. Since

$$\ell'(v) = \frac{Q - nv}{2v^2},$$

we have $\ell'(v) > 0$ for $0 < v < Q/n$ and $\ell'(v) < 0$ for $v > Q/n$. Also, $\ell(v) \to -\infty$ as $v \downarrow 0$ or $v \to \infty$. Therefore Q/n is the unique global maximizer. Hence

$$\boxed{\widehat{\mu}_{\text{MLE}} = \bar{X}}$$

and

$$\boxed{\widehat{\sigma^2}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

Boundary note. If $Q = 0$, all observations are identical. Then the likelihood becomes unbounded as $v \downarrow 0$, so no MLE exists in the parameter space $v > 0$. For a nondegenerate normal population and $n \geq 2$, this event has probability zero.

The variance MLE uses denominator n, not $n - 1$. It is not unbiased:

$$E[\widehat{\sigma^2}_{\text{MLE}}] = \frac{n-1}{n}\sigma^2.$$

---

5. Reference Distributions

5.1 Standard normal distribution

If

$$Z \sim N(0,1),$$

then

$$P(Z > z_\alpha) = \alpha.$$

For a 95% two-sided procedure,

$$z_{0.025} = 1.96.$$

For a 95% one-sided procedure,

$$z_{0.05} = 1.645.$$

5.2 Chi-square distribution

If

$$Z_1, \ldots, Z_\nu \overset{\text{i.i.d.}}{\sim} N(0,1),$$

then

$$\sum_{i=1}^{\nu} Z_i^2 \sim \chi^2_\nu.$$

For a normal random sample,

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}.$$

This exact result requires a normal population.

5.3 Student-t distribution

If

$$Z \sim N(0,1), \qquad V \sim \chi^2_\nu, \qquad Z \perp V,$$

then

$$\boxed{T = \frac{Z}{\sqrt{V/\nu}} \sim t_\nu}.$$

For a normal random sample,

$$\boxed{\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}}.$$

---

6. Confidence Intervals

6.1 Meaning

A $100(1-\alpha)\%$ confidence interval for θ is a random interval $[L, U]$ satisfying

$$P(L \le \theta \le U) = 1 - \alpha.$$

Correct interpretation:

> If the same sampling procedure is repeated many times, approximately $100(1-\alpha)\%$ of the intervals will contain the true parameter.

6.2 General derivation method

Use a pivotal quantity $Q(X_1, \ldots, X_n; \theta)$ whose distribution is known.

1. Find constants a and b such that

$$P(a \le Q \le b) = 1 - \alpha.$$

2. Substitute the expression for Q.
3. Rearrange the inequality so that θ is in the middle.

4. Read off the lower and upper endpoints.

The following derivations should be remembered because they also explain the formulas.

6.3 Two-sided interval for μ: σ known

Assumptions:

- the observations are independent;
- σ is known;
- either the population is normal, or a large-sample normal approximation for $\bar{X}$ is justified.

If the population is normal, then

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

is exact. If a central-limit approximation is used instead, write

$$Z \approx N(0,1),$$

and regard the resulting confidence level as approximate.

For the exact normal case,

$$P\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Substitute Z:

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Multiply by $\sigma/\sqrt{n} > 0$:

$$P\left(-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Rearrange for μ:

$$P\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Therefore

$$\boxed{\mu \in \left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]}.$$

Equivalently,

$$\boxed{\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}.$$

This interval is exact for a normal population and approximate when justified by a large-sample normal approximation.

6.4 Two-sided interval for μ: σ unknown

Assumptions:

- the observations are independent;
- the population is normal;
- σ is unknown.

Use

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Then

$$P\left(-t_{\alpha/2,n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2,n-1}\right) = 1 - \alpha.$$

Rearranging gives

$$P\left(\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

Therefore

$$\boxed{\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}}.$$

Do not replace σ by S and still use 1.96 for a small normal sample.

6.5 Two-sided interval for σ^2

Assumptions:

- the observations are independent;
- the population is normal.

Let $\nu = n-1$. Since

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_\nu,$$

upper-tail notation gives

$$P\left(\chi^2_{1-\alpha/2,\nu} \le Q \le \chi^2_{\alpha/2,\nu}\right) = 1-\alpha.$$

Substitute Q:

$$P\left(\chi^2_{1-\alpha/2,\nu} \le \frac{(n-1)S^2}{\sigma^2} \le \chi^2_{\alpha/2,\nu}\right) = 1-\alpha.$$

All quantities are positive. Taking reciprocals reverses the inequalities:

$$P\left(\frac{1}{\chi^2_{1-\alpha/2,\nu}} \ge \frac{\sigma^2}{(n-1)S^2} \ge \frac{1}{\chi^2_{\alpha/2,\nu}}\right) = 1-\alpha.$$

Reorder the terms and multiply by $(n-1)S^2$:

$$P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,\nu}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\nu}}\right) = 1-\alpha.$$

Therefore

$$\boxed{\sigma^2 \in \left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]}.$$

The larger chi-square critical value is in the lower-bound denominator.

For σ, take square roots of both endpoints.

6.6 One-sided confidence bounds for μ

A one-sided confidence interval has only one finite endpoint.

Lower confidence bound, σ known

We want

$$P(\mu \ge L) = 1-\alpha.$$

Since

$$P(Z \le z_\alpha) = 1-\alpha,$$

we obtain

$$P\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z_\alpha\right) = 1-\alpha.$$

Rearranging gives

$$P\left(\mu \ge \bar{X} - z_\alpha\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha.$$

Thus the $100(1-\alpha)\%$ lower confidence interval is

$$\boxed{\left[\bar{X} - z_\alpha\frac{\sigma}{\sqrt{n}}, \infty\right)}.$$

Upper confidence bound, σ known

Similarly,

$$\boxed{\left(-\infty, \bar{X} + z_\alpha \frac{\sigma}{\sqrt{n}}\right]}.$$

Lower and upper bounds, σ unknown

For a normal population, replace z_α by $t_{\alpha,n-1}$ and σ by S:

$$\boxed{\left[\bar{X} - t_{\alpha,n-1} \frac{S}{\sqrt{n}}, \infty\right)}$$

and

$$\boxed{\left(-\infty, \bar{X} + t_{\alpha,n-1} \frac{S}{\sqrt{n}}\right]}.$$

Which bound should be used?

- If the question asks for evidence that μ is **greater than** a value, use a lower confidence bound.
- If the question asks for evidence that μ is **less than** a value, use an upper confidence bound.
- A 95% one-sided bound uses $z_{0.05}$ or $t_{0.05,n-1}$, not an $\alpha/2$ critical value.

6.7 One-sided confidence bounds for σ^2

Let

$$A = (n-1)S^2, \qquad \nu = n-1.$$

Using upper-tail chi-square notation:

Lower confidence bound

$$\boxed{\left[\frac{A}{\chi^2_{\alpha,\nu}}, \infty\right)}$$

has confidence level $1-\alpha$.

Upper confidence bound

$$\boxed{\left(0, \frac{A}{\chi^2_{1-\alpha,\nu}}\right]}$$

has confidence level $1-\alpha$.

For one-sided bounds on σ, take the square root of the finite endpoint.

6.8 Detailed confidence-interval solution procedure

Step 1: Identify the parameter

State whether the problem asks for μ, σ^2, or σ.

Step 2: State the assumptions

Examples:

- independent observations;
- normal population;
- known or unknown σ.

Step 3: Choose the pivotal distribution

- mean, known σ: standard normal;
- mean, unknown σ, normal population: t_{n-1};
- normal-population variance: χ^2_{n-1}.

Step 4: Write the correct formula before substituting numbers

This reduces arithmetic and notation errors.

Step 5: Find the critical value

Use $\alpha/2$ for a two-sided interval and α for a one-sided bound.

Step 6: Compute the standard error or chi-square numerator

For a mean:

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \frac{s}{\sqrt{n}}.$$

For a variance:

$$(n-1)s^2.$$

Step 7: Compute the endpoint or endpoints

Keep enough decimal places during intermediate calculations.

Step 8: State the final interval and interpretation

Example:

> A 95% confidence interval for the population mean is $[L, U]$.

For a one-sided bound:

> A 95% lower confidence bound for the population mean is L.

6.9 Worked example: two-sided t interval

Suppose

$$n = 16, \qquad \bar{x} = 20, \qquad s = 4,$$

and the population is normal. Find a 95% confidence interval for μ.

Step 1: Identify the method

The population standard deviation is unknown, so use a one-sample t interval with

$$\nu = n - 1 = 15.$$

Step 2: Critical value

$$t_{0.025,15} = 2.131.$$

Step 3: Standard error

$$\frac{s}{\sqrt{n}} = \frac{4}{\sqrt{16}} = 1.$$

Step 4: Margin of error

$$E = t_{0.025,15}\frac{s}{\sqrt{n}} = 2.131(1) = 2.131.$$

Step 5: Endpoints

$$L = 20 - 2.131 = 17.869,$$

$$U = 20 + 2.131 = 22.131.$$

Therefore

$$\boxed{[17.869, 22.131]}.$$

6.10 Worked example: one-sided lower bound

Suppose

$$n = 25, \qquad \bar{x} = 12, \qquad s = 5,$$

and the population is normal. Find a 95% lower confidence bound for μ.

Step 1: Method

Since σ is unknown, use a t bound with

$$\nu = 24.$$

Step 2: Critical value

A 95% one-sided bound has $\alpha = 0.05$, so use

$$t_{0.05,24} \approx 1.711.$$

Step 3: Standard error

$$\frac{s}{\sqrt{n}} = \frac{5}{\sqrt{25}} = 1.$$

Step 4: Lower endpoint

$$L = \bar{x} - t_{0.05,24}\frac{s}{\sqrt{n}} = 12 - 1.711 = 10.289.$$

Therefore the 95% lower confidence interval is

$$\boxed{[10.289, \infty)}.$$

---

7. Hypothesis Testing

7.1 Hypotheses and tails

The null hypothesis is H_0, and the alternative hypothesis is H_1.

Right-tailed test

$$H_0 : \mu \le \mu_0, \qquad H_1 : \mu > \mu_0.$$

Left-tailed test

$$H_0 : \mu \ge \mu_0, \qquad H_1 : \mu < \mu_0.$$

Two-sided test

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \ne \mu_0.$$

The alternative hypothesis determines the tail or tails.

7.2 Test statistics for one population mean

Known σ

$$\boxed{Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}}$$

For a simple null hypothesis $H_0 : \mu = \mu_0$, and for the boundary value $\mu = \mu_0$ in a one-sided composite null hypothesis,

$$Z \sim N(0, 1)$$

when the population is normal. With a justified large-sample approximation, use $Z \approx N(0, 1)$.

Unknown σ

$$\boxed{T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \qquad \nu = n - 1}$$

For a normal population, under $H_0 : \mu = \mu_0$, or at the boundary $\mu = \mu_0$ of a one-sided composite null hypothesis,

$$T \sim t_{n-1}.$$

For the one-sided mean tests in this handout, critical values and p-values are calculated at the boundary value $\mu = \mu_0$.

7.3 Complete hypothesis-test solution procedure

Step 1: Define the parameter

Example:

> Let μ be the population mean burning rate.

Step 2: State H_0 and H_1

Choose the alternative from the wording of the question, not from the observed data.

- "greater than" gives a right-tailed test;
- "less than" gives a left-tailed test;
- "different from" gives a two-sided test.

Step 3: State the significance level

$$\alpha = 0.05$$

unless another value is given.

Step 4: State the assumptions and choose the test

For a one-sample mean test, check:

- observations are independent;
- the population is normal, or a justified large-sample approximation is used;
- whether σ is known.

Then choose Z or t.

Step 5: Write and compute the observed test statistic

Known σ:

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Unknown σ:

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

For a t-test, also state

$$\nu = n - 1.$$

Step 6: Compute the p-value

For a one-sided composite null hypothesis, use the null distribution at the boundary value $\mu = \mu_0$.

Right-tailed:

$$p = P(T \geq t_{\text{obs}}).$$

Left-tailed:

$$p = P(T \leq t_{\text{obs}}).$$

Two-sided with a symmetric null distribution:

$$p = P(|T| \geq |t_{\text{obs}}|) = 2P(T \geq |t_{\text{obs}}|).$$

Use Z instead of T for a Z-test.

Step 7: Make the decision

$$\boxed{p \leq \alpha \Rightarrow \text{reject } H_0}$$

$$\boxed{p > \alpha \Rightarrow \text{fail to reject } H_0}.$$

Step 8: State the conclusion in context

If rejecting:

> Since $p = \cdots < \alpha$, reject H_0. There is sufficient evidence at the $100\alpha\%$ significance level that ...

If not rejecting:

> Since $p = \cdots > \alpha$, fail to reject H_0. There is insufficient evidence at the $100\alpha\%$ significance level to conclude that ...

Do not write “accept H_0.”

7.4 Critical-value method

A test may also be solved using a rejection region.

Right-tailed

Reject H_0 if

$$Z \geq z_\alpha$$

or

$$T \geq t_{\alpha,n-1}.$$

Left-tailed

Reject H_0 if

$$Z \leq -z_\alpha$$

or

$$T \leq -t_{\alpha,n-1}.$$

Two-sided

Reject H_0 if

$$|Z| \geq z_{\alpha/2}$$

or

$$|T| \geq t_{\alpha/2,n-1}.$$

The p-value method and the critical-value method give the same decision when used correctly. For the continuous reference distributions used here, equality at a critical value has probability zero.

7.5 Detailed worked example: two-sided Z-test

A process is designed to have mean burning rate 40. Assume that the observations are independent and come from a normal population. Suppose

$$\sigma = 2, \qquad n = 25, \qquad \bar{x} = 41.25.$$

Test at $\alpha = 0.05$ whether the population mean differs from 40.

Step 1: Parameter

Let μ be the population mean burning rate.

Step 2: Hypotheses

$$H_0 : \mu = 40, \qquad H_1 : \mu \neq 40.$$

This is a two-sided test.

Step 3: Test selection

The population standard deviation is known and the population is assumed normal, so use an exact one-sample Z-test.

Step 4: Test statistic

$$\begin{aligned} z_{\text{obs}} &= \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \\ &= \frac{41.25 - 40}{2/\sqrt{25}} \\ &= \frac{1.25}{0.4} \\ &= 3.125. \end{aligned}$$

Step 5: p-value

Because the test is two-sided,

$$
\begin{aligned}
p &= 2P(Z \geq |3.125|) \\
&\approx 2(0.0009) \\
&= 0.0018.
\end{aligned}
$$

Step 6: Decision

$$0.0018 < 0.05.$$

Therefore, reject H_0.

Step 7: Conclusion

> There is sufficient evidence at the 5% significance level that the population mean burning rate differs from 40.

7.6 Detailed worked example: right-tailed t-test

Suppose

$$n = 16, \qquad \bar{x} = 52, \qquad s = 8.$$

Assume a normal population. Test at $\alpha = 0.05$:

$$H_0 : \mu \leq 48, \qquad H_1 : \mu > 48.$$

Step 1: Test selection

The population standard deviation is unknown, so use a one-sample t-test. For the one-sided composite null, calculate the null distribution at the boundary value $\mu = 48$. The degrees of freedom are

$$\nu = n - 1 = 15.$$

Step 2: Test statistic

$$
\begin{aligned}
t_{\text{obs}} &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \\
&= \frac{52 - 48}{8/\sqrt{16}} \\
&= \frac{4}{2} \\
&= 2.
\end{aligned}
$$

Step 3: p-value

This is a right-tailed test:

$$p = P(T_{15} \geq 2) \approx 0.032.$$

Step 4: Decision

$$0.032 < 0.05.$$

Reject H_0.

Step 5: Conclusion

> There is sufficient evidence at the 5% significance level that the population mean is greater than 48.

7.7 Confidence intervals and tests

For matching two-sided procedures, apart from the probability-zero equality case at a continuous critical value,

$$\boxed{\mu_0 \notin \text{the } 100(1-\alpha)\% \text{ confidence interval} \iff H_0 : \mu = \mu_0 \text{ is rejected at level } \alpha}.$$

For one-sided procedures:

- a right-tailed test $H_1 : \mu > \mu_0$ corresponds to a lower confidence bound;
- a left-tailed test $H_1 : \mu < \mu_0$ corresponds to an upper confidence bound.

For example, in a right-tailed level-α test, reject $H_0 : \mu \leq \mu_0$ exactly when the $100(1-\alpha)\%$ lower confidence bound is greater than μ_0.

7.8 Meaning of the p-value

For a simple null hypothesis, the p-value is

$$P(\text{a result at least as extreme as observed} \mid H_0).$$

For the one-sided composite mean hypotheses used here, calculate this probability at the boundary value $\mu = \mu_0$.

It is not

$$P(H_0 \mid \text{data}).$$

A small p-value means that the data are difficult to explain under H_0. It does not measure the size or practical importance of an effect.

---

8. Formula Sheet

8.1 Basic sample quantities

$$\boxed{\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i}$$

$$\boxed{S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$$

$$\boxed{S = \sqrt{S^2}}$$

$$\boxed{S^2 = \frac{n\sum_i X_i^2 - (\sum_i X_i)^2}{n(n-1)}}$$

8.2 Sampling properties

$$\boxed{E[\bar{X}] = \mu}$$

$$\boxed{\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}}$$

$$\boxed{\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}}$$

$$\boxed{\widehat{\mathrm{SE}}(\bar{X}) = \frac{S}{\sqrt{n}}}$$

$$\boxed{E[S^2] = \sigma^2}$$

8.3 Method of moments

$$\boxed{M_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k}$$

Assuming the required moments exist and identify the parameters, replace population moments by sample moments and solve.

8.4 MLE exam chain

$$\boxed{L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)}$$

$$\boxed{\ell(\theta) = \log L(\theta)}$$

$$\boxed{\ell'(\hat{\theta}) = 0}$$

This gives a candidate. Then prove that it is a **global** maximizer by strict concavity on the parameter space, a positive-to-negative score sign change, comparison with all candidates and boundaries, or a direct likelihood argument. The condition

$$\ell''(\hat{\theta}) < 0$$

by itself proves only a strict local maximum.

Always check the parameter space and boundaries.

Important results:

$$\boxed{X_i \sim \mathrm{Poisson}(\lambda) \implies \widehat{\lambda}_{\mathrm{MLE}} = \bar{X}}$$

$$\boxed{X_i \sim N(\mu, \sigma^2) \implies \widehat{\mu}_{\mathrm{MLE}} = \bar{X}}$$

$$\boxed{\widehat{\sigma^2}_{\mathrm{MLE}} = \frac{1}{n}\sum_i (X_i - \bar{X})^2}$$

8.5 Reference pivots

For a normal sample,

$$\boxed{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)}$$

For a justified large-sample approximation with a nonnormal population, replace $\sim$ by $\approx$.

$$\boxed{\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}}$$

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}$$

8.6 Two-sided confidence intervals

Mean, σ known

$$\boxed{\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}$$

Mean, σ unknown

$$\boxed{\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}}$$

Variance

$$\boxed{\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]}$$

For σ, take square roots.

8.7 One-sided confidence bounds

Mean, known σ

Lower bound:

$$\boxed{\left[\bar{X} - z_\alpha\frac{\sigma}{\sqrt{n}}, \infty\right)}$$

Upper bound:

$$\boxed{\left(-\infty, \bar{X} + z_\alpha\frac{\sigma}{\sqrt{n}}\right]}$$

Mean, unknown σ

Lower bound:

$$\boxed{\left[\bar{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}}, \infty\right)}$$

Upper bound:

$$\boxed{\left(-\infty, \bar{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}\right]}$$

Variance

Let $A = (n-1)S^2$ and $\nu = n-1$.

Lower bound:

$$\boxed{\left[\frac{A}{\chi^2_{\alpha,\nu}}, \infty\right)}$$

Upper bound:

$$\boxed{\left(0, \frac{A}{\chi^2_{1-\alpha,\nu}}\right]}$$

8.8 One-sample mean test statistics

Known σ:

$$\boxed{Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}}$$

Unknown σ:

$$\boxed{T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \qquad \nu = n - 1}$$

8.9 p-values

Right-tailed:

$$\boxed{p = P(T \geq t_{\text{obs}})}$$

Left-tailed:

$$\boxed{p = P(T \leq t_{\text{obs}})}$$

Two-sided:

$$\boxed{p = 2P(T \geq |t_{\text{obs}}|)}$$

Use Z instead of T for a Z-test.

8.10 Critical-value rejection regions

Alternative	Z-test rejection region	t-test rejection region
$H_1 : \mu > \mu_0$	$Z \geq z_\alpha$	$T \geq t_{\alpha,n-1}$
$H_1 : \mu < \mu_0$	$Z \leq -z_\alpha$	$T \leq -t_{\alpha,n-1}$
$H_1 : \mu \neq \mu_0$	$\lvert Z\rvert \geq z_{\alpha/2}$	$\lvert T\rvert \geq t_{\alpha/2,n-1}$

8.11 Decision rule

$$\boxed{p \leq \alpha \Rightarrow \text{reject } H_0}$$

$$\boxed{p > \alpha \Rightarrow \text{fail to reject } H_0}$$

8.12 MLE workflow

1. State the parameter space.
2. Write $L(\theta)$.
3. Write $\ell(\theta)$.
4. Solve $\ell'(\theta) = 0$ to obtain candidate points.
5. Prove that the candidate is the global maximizer. Use strict concavity on the parameter space, a positive-to-negative score sign change, comparison with all candidates and boundaries, or a direct likelihood argument.
6. Remember that $\ell''(\hat{\theta}) < 0$ alone proves only a local maximum.
7. Check boundaries and parameter restrictions.
8. State the MLE clearly.

8.13 Confidence-interval workflow

1. Identify the parameter.
2. State the assumptions.
3. Decide whether the interval is two-sided, lower one-sided, or upper one-sided.
4. Choose Z, t, or χ^2.
5. Use $\alpha/2$ for a two-sided interval and α for a one-sided bound.
6. Write the formula before substituting values.
7. Compute the endpoint or endpoints.

8. If estimating σ, take square roots of a variance interval.
9. State the answer in context.

8.14 Hypothesis-test workflow

1. Define the parameter.
2. State H_0 and H_1.
3. State α.
4. State assumptions and choose Z or t. Distinguish an exact normal model from a large-sample approximation.
5. For a one-sided composite null, use the boundary value $\mu = \mu_0$.
6. Compute the observed test statistic.
7. Use H_1 to choose the correct tail or tails.
8. Compute the p-value.
9. Compare p with α.
10. State “reject” or “fail to reject.”
11. Write the conclusion in the context of the problem.